From the Mutual Fund Theorem to the Security Market Line

Does the theory survive real data?

FE 5108 Portfolio Theory and Investments

Group members and contributions

Member	Student ID	Contribution
Rahul Thaker	A0294733W	Project scaffold and reproducible pipeline; point-in-time market caps and value weights; US and Singapore data loaders, risk-free proxies and the SORA build; Stage 1 portfolio choice, estimation-error analysis and no-short/shrinkage robustness; Stage 2 CAPM and empirical SML; Stage 3 size and momentum factor construction and the factor-model comparison; paired-market visualisation; Stage 4 memo and out-of-sample validation; report writing, editing and PR integration.
Yi-Kai Tsai	A0355345B	The robustness analysis: built the opt-in Newey-West (HAC) covariance machinery and used it to re-test every Stage-2 and Stage-3 alpha at 0, 3 and 6 monthly lags, quantifying how the standard-error choice changes which individual alphas survive; integrated the robustness results into the report and reviewed the draft against the project requirements.
Aryan Sethi	A0330514U	Singapore data cleaning: rebuilt every SGX total-return series and point-in-time market cap from the daily cache, the SGX corporate actions and month-end USD/SGD, documenting each correction with its source; built and compared the custom factors across both markets, including the industry-momentum factor on 60 years of Ken French US industries with its premium, alphas, turnover and break-even cost.
Deepank Dewan	A0357157W	Stage 3 factor support: built the low volatility (LOWVOL) and reversal (REV) signals and their spread portfolios, including turnover and break-even cost analysis; ran the factor comparison across both markets and maintained the US and Singapore factor analysis notebook; reviewed the final report.

The theory makes two claims on the same data: a recipe for the best portfolio, and a description of how risk is priced. We test both on 30 US assets and 29 Singapore assets over 119 months. The main finding is about the sample: it holds the decade's winners and none of its failures, so the value-weighted index we are benchmarked against is the survivors' own portfolio. Every alternative is tested on that field, and the field drives the conclusion.

Contents

- Executive summary
- Data and methodology
- Stage 1: portfolio choice and estimation error
- Stage 2: the CAPM and the empirical SML
- Stage 3: factors
- Stage 4: the memo to the investment committee
- Appendix

Executive summary

The main finding is about the data, not the theory. Both universes are today's index members carried back to 2016, so the failed firms are absent, the largest survivors compounded, and the value-weighted index is the survivors' own portfolio. A tilt away from cap weights is a bet against those winners, which makes beating the index on this universe unreasonable at the outset.

On that field the CAPM's risk pricing looks reasonable in the US (2 of 30 alphas significant, GRS p = 0.47, fitted slope 0.016 against theory's 0.019) and fails in Singapore (slope 0.001 against 0.011, R squared 0.004). Both fits are computed on survivors, so the US pass is flattering rather than clean. The portfolio recipe fails in both markets for a separate reason, estimation error: the tangency portfolio shorts the consumer-staples ETF by 280% of capital and takes 16.3x gross exposure, its half-sample weights correlate at 0.35 (US) and -0.01 (Singapore), and in the monthly out-of-sample test the index still wins (US Sharpe 1.10 against 0.45 for the raw tangency and 0.83 for the best estimated rule; Singapore 1.77 against 0.50 and 1.18).

Factors add explanatory power without absorbing the alphas: the US mean |alpha| moves from 0.45% to 0.52% a month and Singapore's from 0.63% to 0.71%, with stars from 5 to 8. The sorts have no true loser leg, because survivorship deleted it. The clean test is the survivor-free industry momentum study, where momentum is priced (3.8% a year, t = 2.2) but breaks even at 38 basis points of trading cost.

Judgment. Hold the value-weighted index, or at most a shrunk long-only tilt. The CAPM verdict is also proxy-dependent: swapping our index for the Fama-French factor halves the US GRS p value (0.47 to 0.19) and flips the fitted line from too flat to too steep. Whether active management can beat the index in general is a question this sample cannot answer.

Data and methodology

We use 30 US securities (21 mega-caps and 9 sector ETFs) and 29 Singapore listings, 119 monthly returns from October 2016 to August 2026. US returns are Yahoo adjusted closes; Singapore returns are rebuilt from raw closes, dividends and SGX corporate actions, because Yahoo's adjusted close is wrong for five names (Appendix). The US risk-free rate is the Fama-French one-month bill, and Singapore uses the MAS compounded SORA index. The market proxy is a value-weighted portfolio of the same universe, replaced by the Fama-French factor in the Stage 4 Roll check. Stage 3 builds size, momentum, low-volatility and reversal spreads from each universe.

The main finding starts with the sample. Both panels are today's index members carried back to 2016, so failed and acquired firms are missing, average returns and alphas are biased upward, and the index is the survivors' own portfolio. That

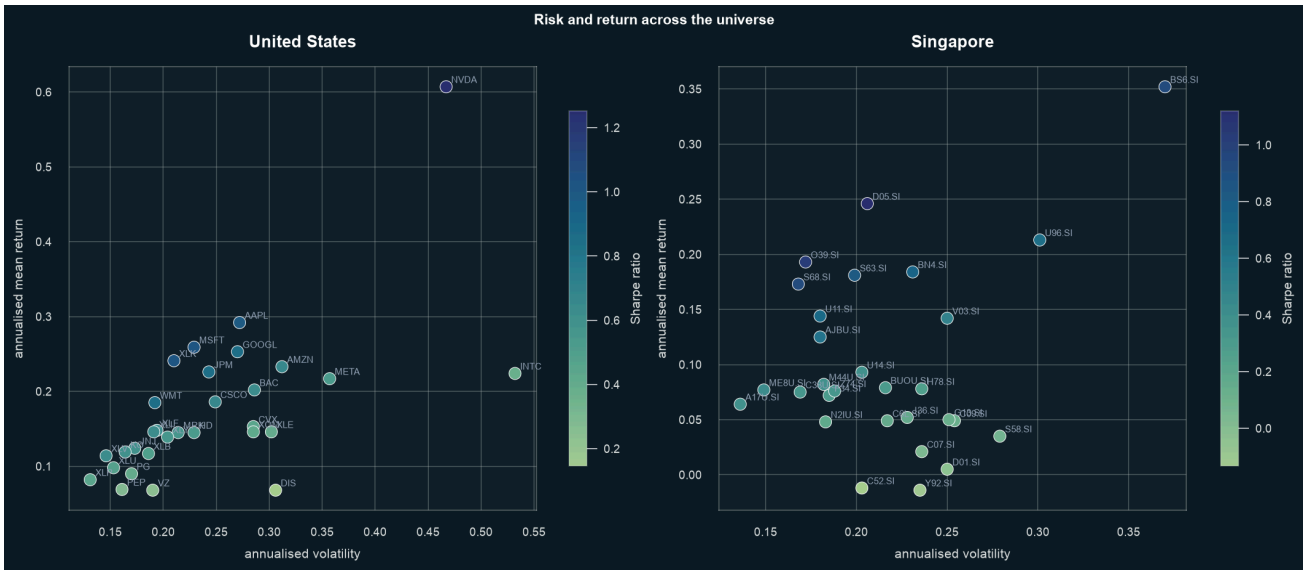
limits what Stages 1 to 3 can conclude, and it is why Stage 4 leans on the survivor-free industry test.

The sample. Both panels are balanced, 119 months each, so every comparison runs on one calendar. They are also survivor panels: today's index members, with the failures missing and the benchmark built from survivors rather than the historical S&P 500 or STI.

Stage 1: portfolio choice and estimation error

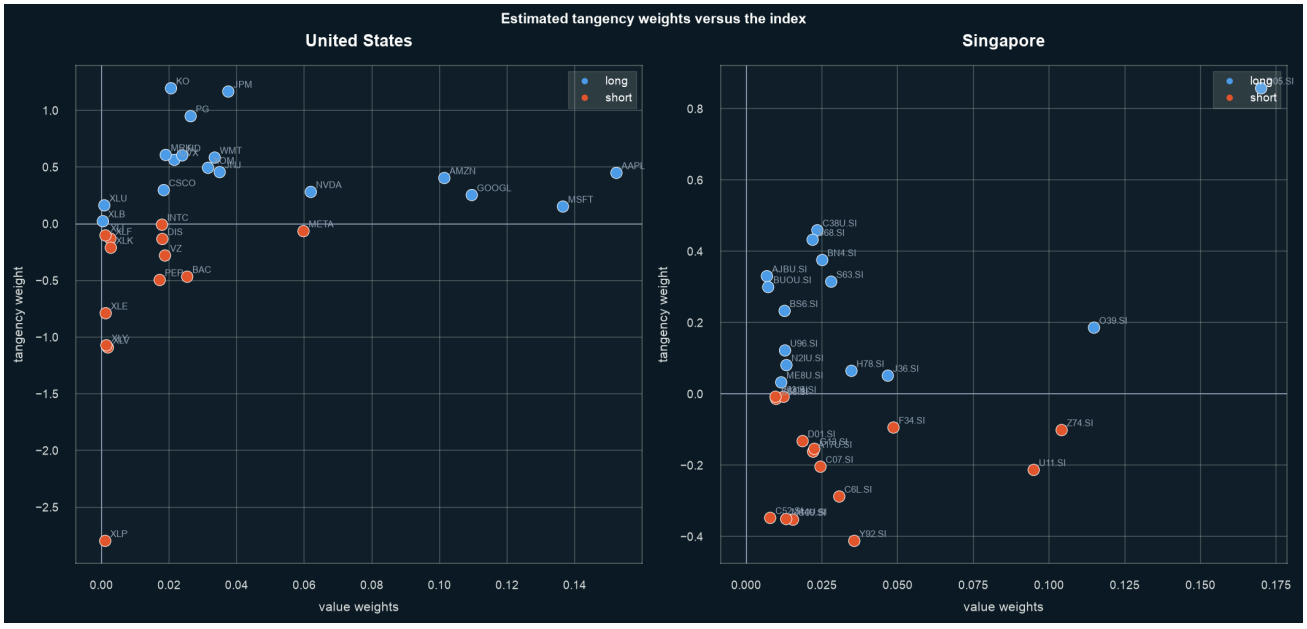
The Mutual Fund Theorem is an identity that takes its inputs as given. We estimate them from 119 months and compare the result with the value-weighted index, which needs no estimates but is itself the survivors' portfolio.

(1a) Summary statistics



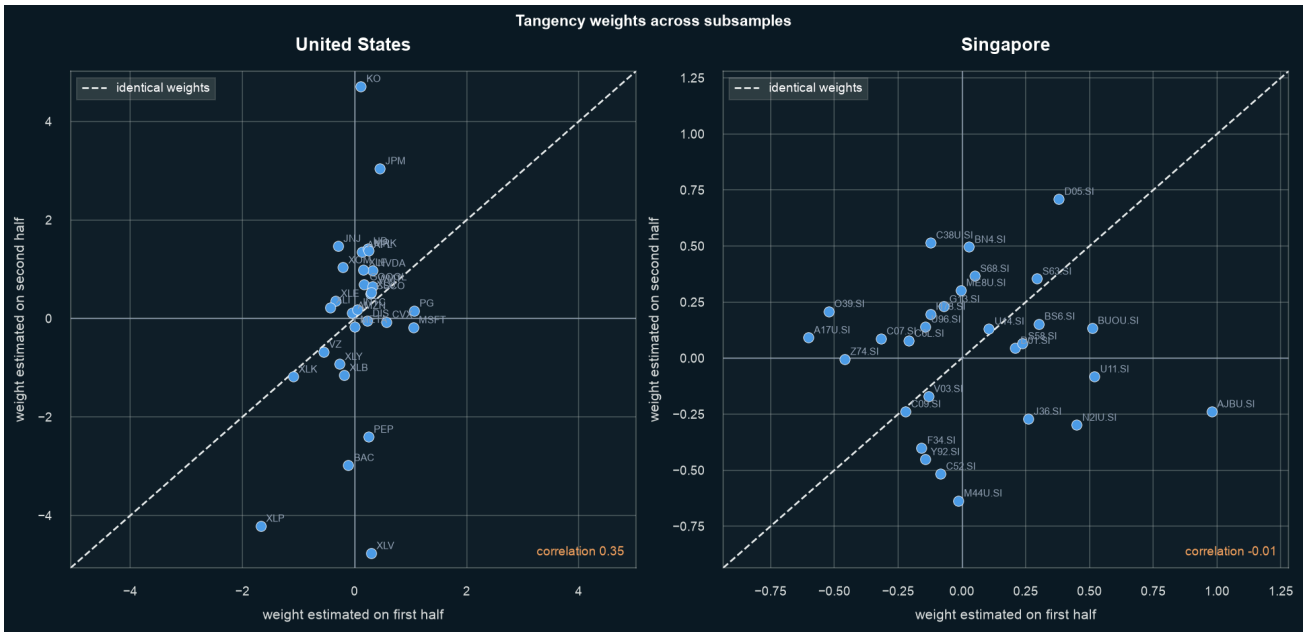
Takeaway. Means are dispersed (US 7% DIS to 61% NVDA; Singapore -1% Y92 to 35% BS6) while volatilities are not (13% to 53% US, 14% to 37% Singapore), and the means are the shakier input: across halves the average US mean moves about 10 percentage points against 5 for volatility. An optimiser fed these means is trading its estimation window.

(1b) Tangency weights versus value weights



Takeaway. Inverting the covariance matrix turns estimation error into positions: the tangency portfolio shorts XLP by 280%, takes 16.3x gross exposure, and bears no resemblance to the index, which holds 11% to 23% in its largest name.

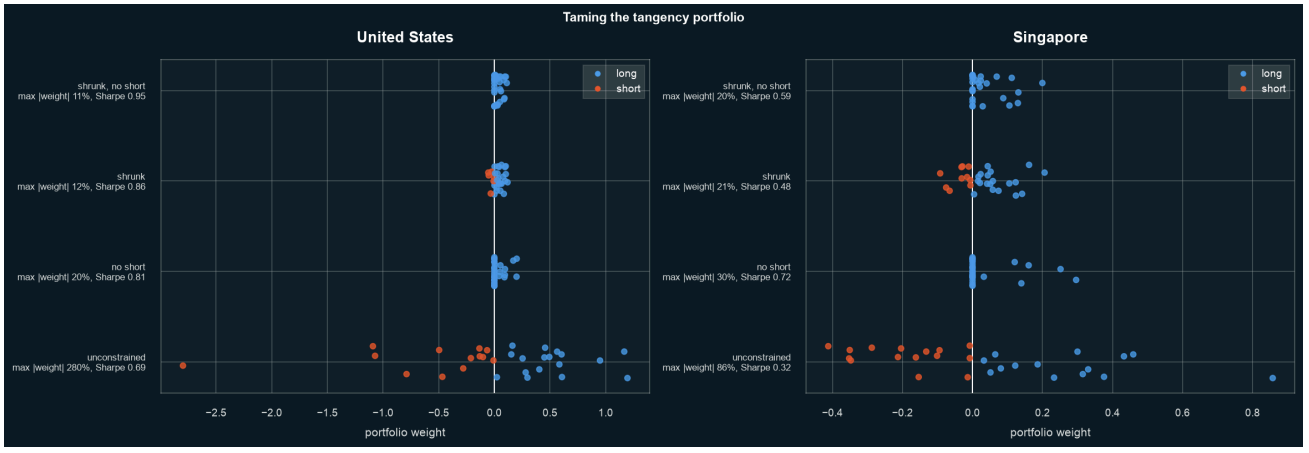
(1c) How stable are the weights?



Takeaway. The two half-sample portfolios barely correlate (0.35 in the US, -0.01 in Singapore). XLV moves from 0.29 to -4.76 and KO from 0.11 to 4.72; a weight that volatile is noise, not a signal.

Taming the tangency portfolio

A no-short constraint and shrinkage of the covariance and the means reduce the extremes. The strip plot scores four versions.



Takeaway. The no-short constraint caps the largest US position at 20% and removes all 13 shorts; adding shrinkage brings it to 11% with leverage 1.0x against 16.3x raw, and the split-half Sharpe rises from 0.69 to 0.95. The fixes remove positions no committee would approve, but the benchmark is still the survivors' portfolio; Stage 4 tests whether they beat it.

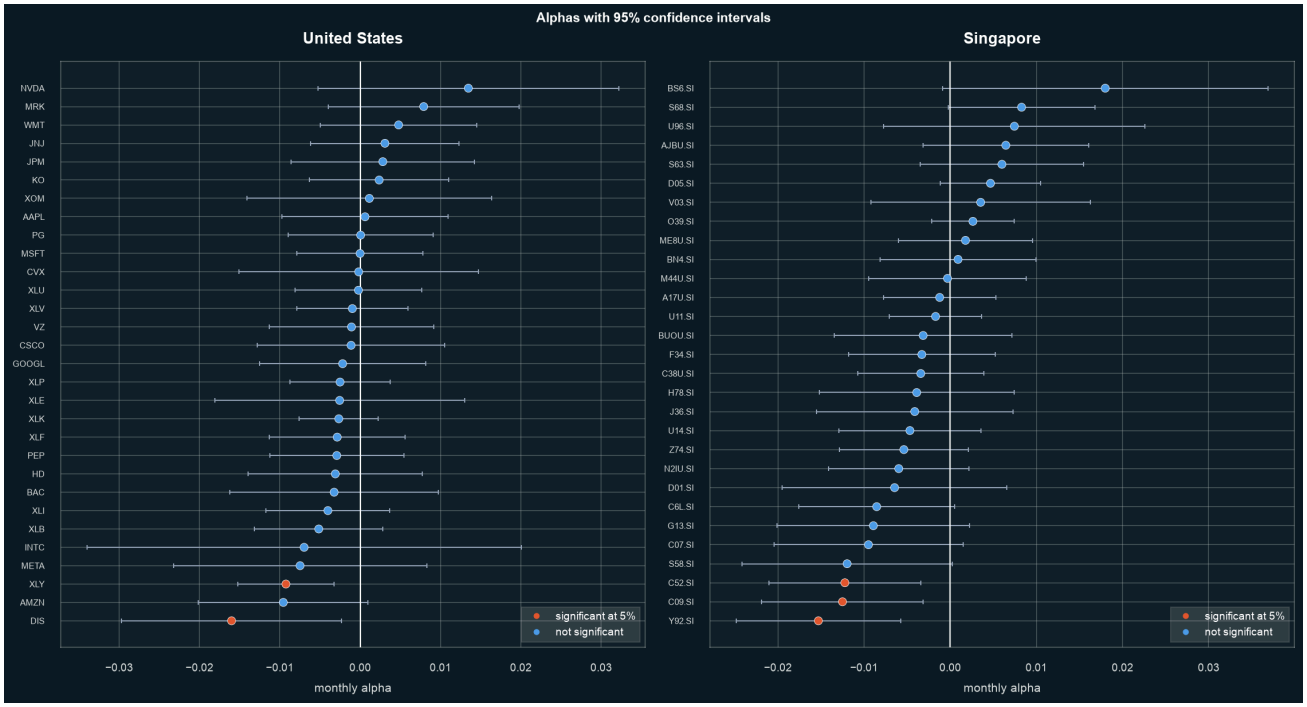
Stage 2: the CAPM and the empirical SML

Stage 1 asked what to hold. Stage 2 asks whether average returns line up with risk:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i \left(R_{M,t} - R_{f,t}\right) + \varepsilon_{i,t},$$

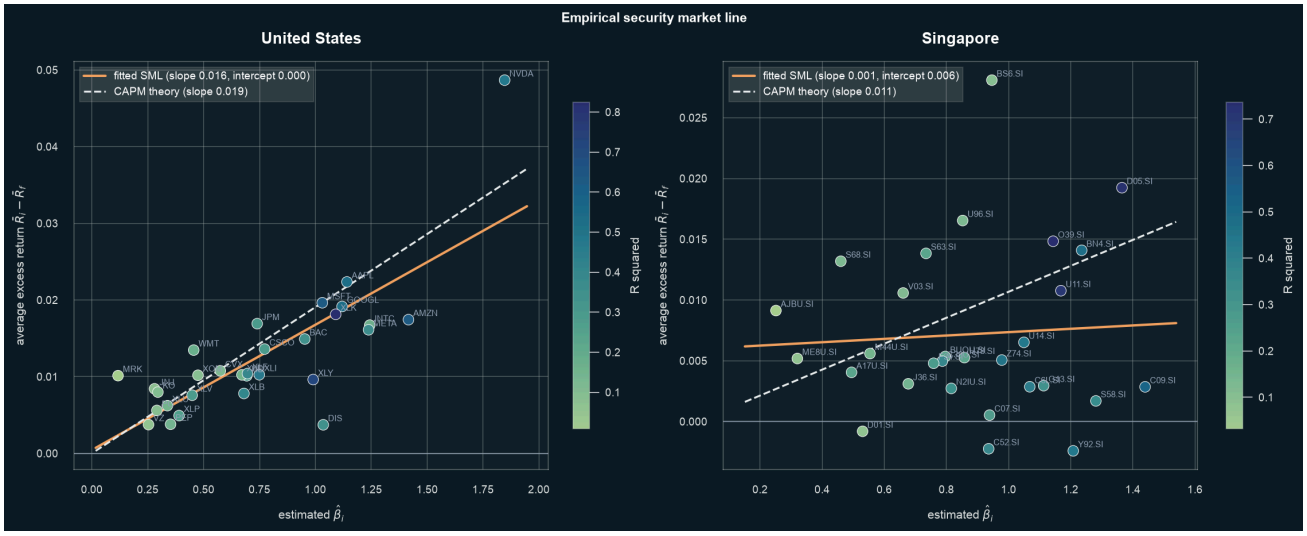
with all alphas zero and expected excess return proportional to beta. One or two values of |t(alpha)| above 2 are expected by chance, so the joint test and the fitted line carry the argument. Both are estimated on survivor panels.

(2a) Security characteristic lines



Takeaway. Betas are precise (US range 0.12 to 1.85, mean R squared 0.31) and alphas are not: only DIS and XLY clear |t| = 2 in the US, against 1.5 expected by chance, and 3 of 29 clear it in Singapore. The full per-asset panel is in the data files.

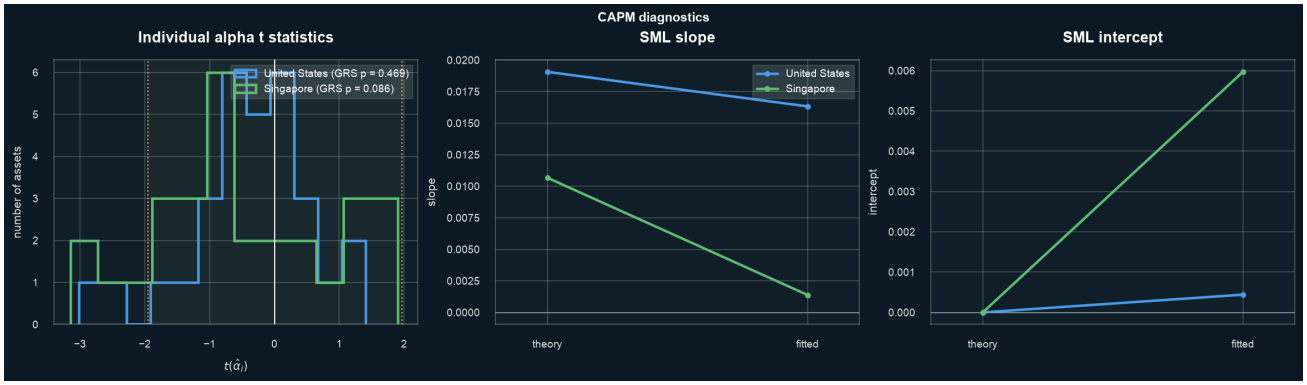
(2b) The empirical SML



Takeaway. The US line is close to theory (slope 0.016 against 0.019, intercept 0.000, R squared 0.61). Singapore's is flat (0.001 against 0.011, intercept 0.006, R squared 0.004), so beta organises US returns and not Singapore's. Both lines are drawn on survivor scatters.

(2c–2d) Are the alphas zero, and does the SML fit?

The GRS p value is the probability of seeing alphas this large if every true alpha were zero. **The fitted SML** regresses average excess return on estimated beta; theory requires intercept zero and slope equal to the average market excess return.



Takeaway. The US passes both diagnostics (2 stars of 30, GRS p = 0.47). Singapore fails the fitted line badly (slope 0.011 to 0.001, intercept 0.006) while the joint test is only marginal (p = 0.09): the test lacked power, the line did not fit. Neither verdict can settle the CAPM, because the cross-section that would decide it includes the firms that did not survive.

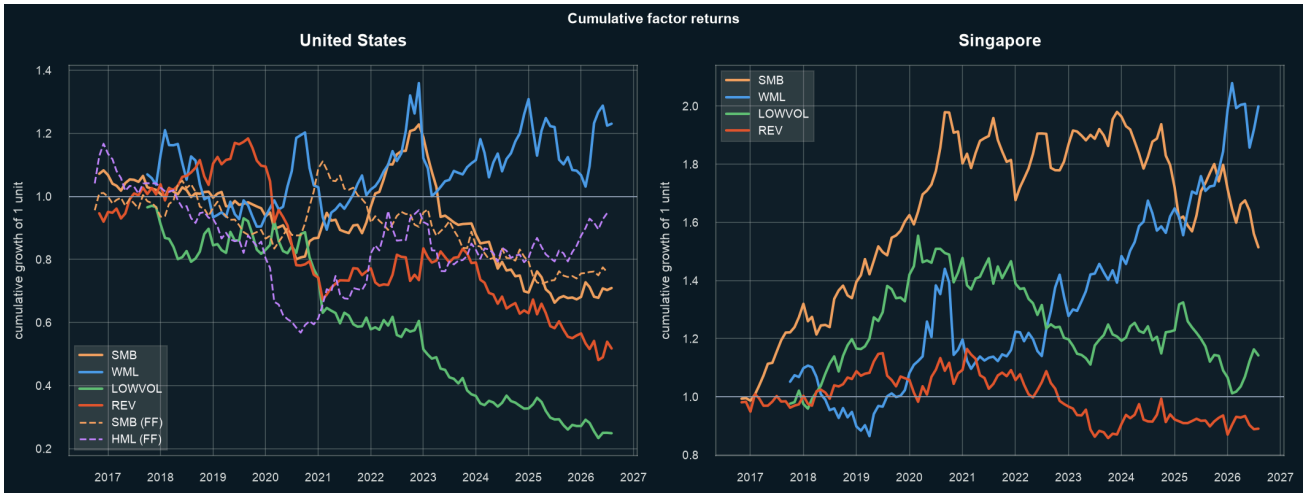
Stage 3: factors

Stage 3 adds size, momentum, low volatility and reversal. The US has FF3 and momentum downloads as benchmarks; Singapore has no library, so all four factors are built from the universe. Every model uses the same 106 or 107 months, so the Stage 3 CAPM baseline (5 Singapore stars) is not the Stage 2 headline (3). The sorts also have no true loser leg: survivorship deleted it.

(3a) How the factors are built

- **Size (SMB):** long the small-cap third, short the large-cap third, sorted on lagged point-in-time value weights.
- **Momentum (WML):** long past winners, short past losers, ranked on the return from month t-12 to t-2.
- **Low volatility (LOWVOL):** long the calmest third, short the most volatile, ranked on trailing 12-month volatility.
- **Reversal (REV):** long last month's losers, short its winners.

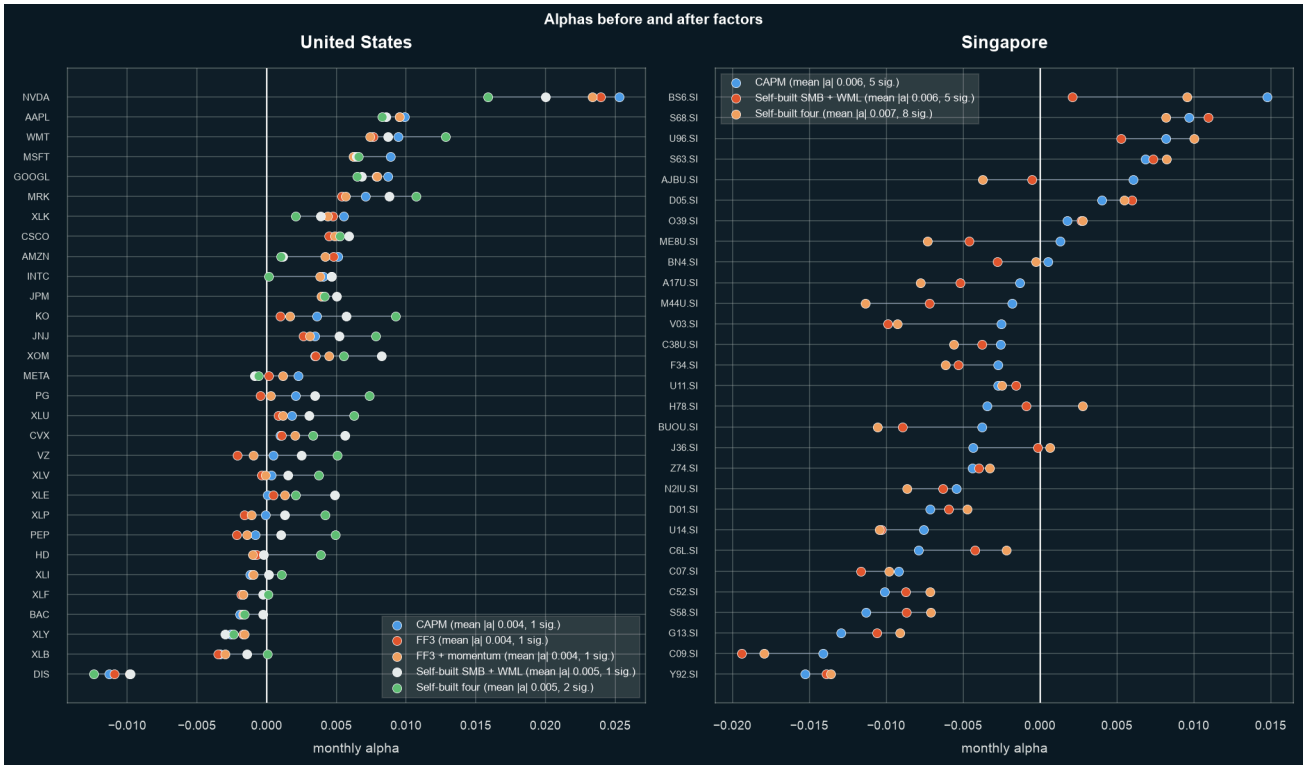
All are equal-weighted 30/40/30 spreads built from the same returns they help explain, and from survivors only.



Takeaway. US benchmarks were subdued over the decade (FF size 0.76x, value 0.57x at its 2020 low) and our self-built US factors worse (size 0.71x, reversal 0.52x, low volatility 0.25x). Singapore's momentum doubles and size gains half, but the "small" leg is another large STI name. None of the spreads is a stable standalone return source.

(3b) Alphas before versus after factors

Each asset's alpha under every model on the common sample, ordered by its CAPM alpha. The legend carries each model's mean |alpha| and star count; all models in a market share the same market factor and months.



Takeaway. The factors add explained variance (US mean R squared 0.38 to 0.55) without shrinking the alphas: US mean |alpha| rises from 0.45% to 0.52% a month, Singapore's from 0.63% to 0.71% with stars from 5 to 8. In Singapore the joint test sharpens from 0.07 to 0.004 because the factors shrink the residual covariance, not the mispricing. Explanatory power is not anomaly absorption.

(3c) Which anomalies survive?

The table lists every asset whose alpha is significant before or after the factors (`|t| > 2`).

	asset	CAPM alpha	CAPM t	Self-built four alpha	Self-built four t
US	KO	0.0036	0.7939	0.0093	2.5610
	NVDA	0.0253	2.4032	0.0159	1.7398
	WMT	0.0094	1.8261	0.0128	2.6820
SG	A17U.SI	-0.0013	-0.3654	-0.0078	-2.4322
	BUOU.SI	-0.0038	-0.6594	-0.0105	-2.0109
	C09.SI	-0.0141	-2.7365	-0.0179	-3.6068
	C52.SI	-0.0101	-2.1852	-0.0072	-1.6411
	G13.SI	-0.0130	-2.4149	-0.0091	-1.6376
	M44U.SI	-0.0018	-0.3547	-0.0114	-2.7133
	ME8U.SI	0.0013	0.3076	-0.0073	-2.0991
	N2IU.SI	-0.0055	-1.2233	-0.0086	-2.1558
	S68.SI	0.0097	2.0371	0.0082	1.6722
	U14.SI	-0.0076	-1.6905	-0.0104	-2.3456
	Y92.SI	-0.0153	-3.0149	-0.0136	-2.5398

Verdict. In the US the factors absorb NVDA (2.5% to 1.6% a month, t 2.40 to 1.74) but KO and WMT become significant, so the star count rises and no model rejects jointly (GRS p = 0.27 against 0.46). In Singapore five stars become eight, mostly negative REITs; C52, G13 and S68 drop out, A17U, BUOU, M44U, ME8U, N2IU and U14 appear, and C09 and Y92 stay negative. The joint test sharpens from 0.07 to 0.004. Neither result speaks to factor premia: both are computed only on survivors.

Stage 4: the memo to the investment committee

The recommendation and the out-of-sample test are the memo; the checks that follow - the factor-hunt argument, the survivor-free industry test and Roll's critique - are the evidence behind it.

The recommendation: hold the index

The theorem says how to build the best portfolio, not what it is. Its answer needs a mean vector and a covariance matrix, both estimated from a finite sample, and the tangency portfolio inverts them: it bets most where the estimates are least stable, and it shorts the winners (XLP by 280%) in a universe that contains no failed firms to find value in. Re-estimated

monthly and traded forward it is the weakest rule in both markets and carries the deepest drawdown, while the value-weighted index is the strongest. The index is the survivors' own portfolio, so that ranking is partly set by the sample before any estimation begins.

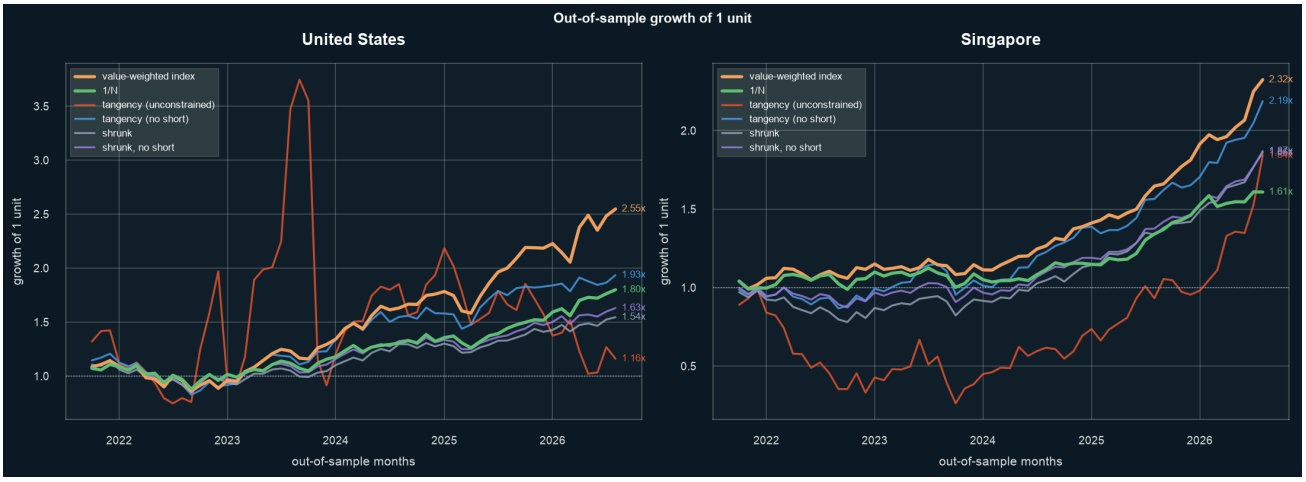
We would not fund the tangency portfolio: our estimates cannot support those positions, and this universe gives an active strategy no obvious room to beat the index. Hold the value-weighted index, or at most a shrunk long-only tilt. Whether active management can beat the index in general is a question this sample cannot answer.

Out-of-sample validation

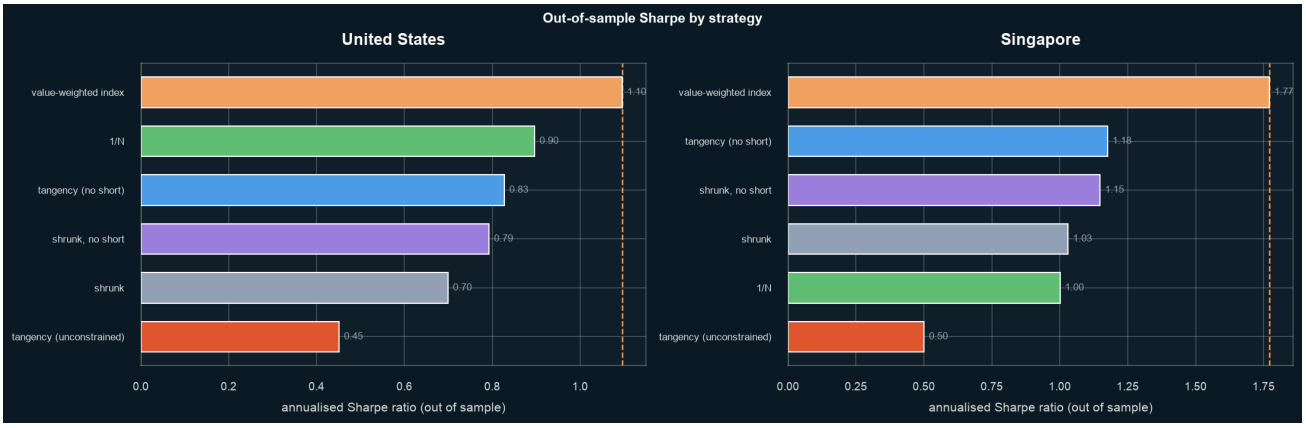
Each strategy is re-estimated every month on an expanding window and held for one month, so no weight ever sees the return it is traded on. The first 60 months train and the remaining 59 are scored; the index and 1/N need no estimation.

		months	ann. mean	ann. vol	Sharpe	max drawdown
us	**index**	59.0	0.210	0.191	1.096	-0.246
	equal_weight	59.0	0.130	0.146	0.896	-0.206
	unconstrained	59.0	0.339	0.752	0.451	-0.756
	long_only	59.0	0.151	0.183	0.828	-0.314
	shrunk	59.0	0.098	0.141	0.699	-0.263
	shrunk_long_only	59.0	0.109	0.138	0.793	-0.241
sg	**index**	59.0	0.178	0.100	1.771	-0.082
	equal_weight	59.0	0.102	0.102	1.001	-0.109
	unconstrained	59.0	0.262	0.524	0.500	-0.737
	long_only	59.0	0.171	0.145	1.176	-0.171
	shrunk	59.0	0.136	0.132	1.030	-0.204
	shrunk_long_only	59.0	0.133	0.116	1.148	-0.127

Takeaway. The index wins on Sharpe in both markets (1.10 in the US, 1.77 in Singapore) and the raw tangency is last (0.45 and 0.50) with drawdowns of -76% and -74%. The no-short rule is the best estimated variant (0.83 and 1.18), and even 1/N beats the estimated rules in the US. The index's lead is the survivor construction working: a tilt away from the largest survivors gives up the concentration that paid.



Takeaway. Over the 59 months the index compounds to 2.55x in the US and 2.32x in Singapore; the raw tangency ends at 1.16x in the US after a spike to 3.75x. No estimated rule beats the survivors' portfolio once risk is priced, and that gap is the cost of estimating inputs rather than inheriting them.



Verdict. The index is the rule to hold: no estimated construction beats it on Sharpe in either market, and the ranking is partly the sample's doing.

The main finding: this universe decides the answer

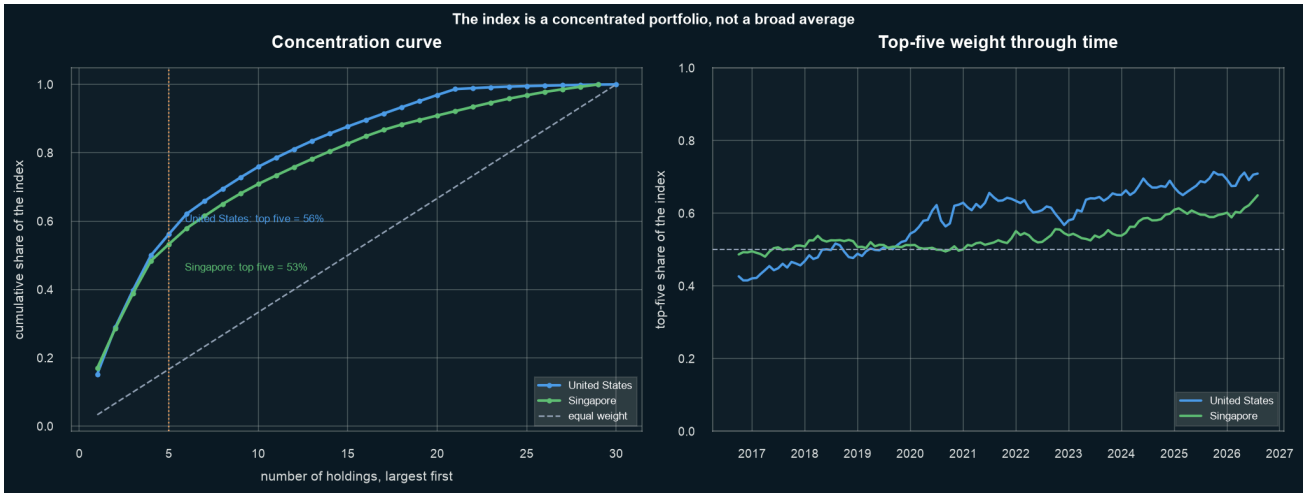
This is the report's central result. A weak factor finding invites a hunt for more factors; the problem is the instrument, not the search.

The index is the concentrated bet. The US universe is 21 mega-caps plus nine sector ETFs and Singapore is roughly 29 large STI names. The sample-average top five weights are 56% and 53%, the top-five share is persistent, and over this decade the leaders compounded. Any tilt away from cap weights is a bet against them.

Survivorship pre-loads the winners. Both panels are today's members carried back to 2016, so failed firms are absent. A factor sort needs a loser leg, and we have deleted the losers: measured spreads are compressed or reversed, and the Singapore size factor's "small" leg is another large STI name.

Thirty names is too few bets. Re-formed monthly over 119 months, the panel gives a few dozen meaningful bets per leg: enough to see the market factor and the SML, not enough to resolve a premium of a fraction of a percent a month.

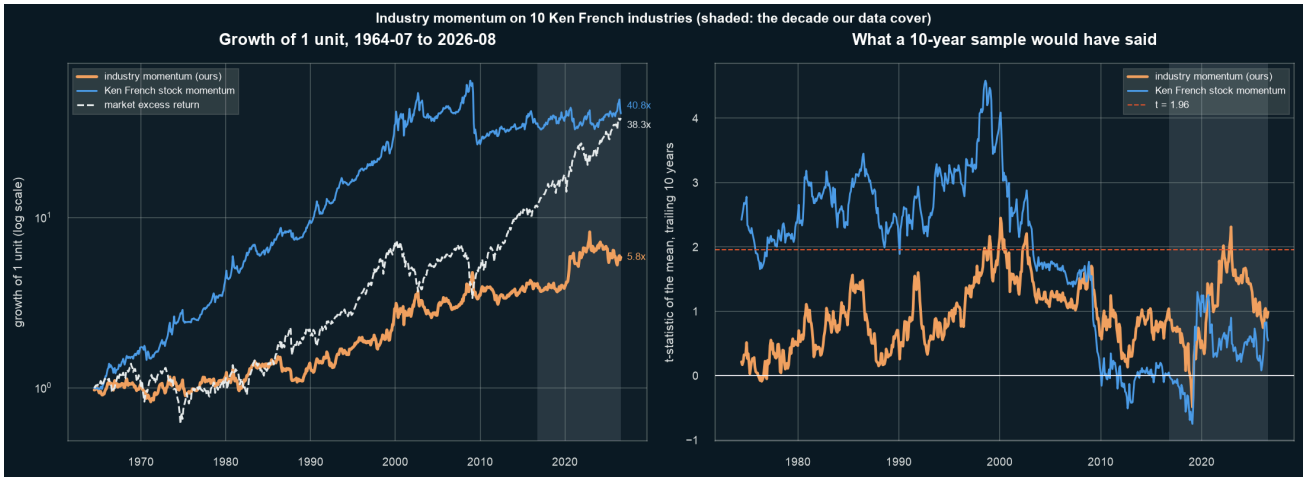
This universe is the wrong instrument for testing factor premia. Momentum is the one factor we can rerun on a survivor-free instrument, and we do it next.



Takeaway. The "market" is a concentrated portfolio: the largest name carries 15% of the US index (17% in Singapore) and the top five 56% and 53%, a share that rises through the sample (US 43% to 71%). The index is the concentrated portfolio of survivors, and no tilt here can diversify away from the names that made it.

Industry momentum: the survivor-free test

Ken French's ten value-weighted US industry portfolios hold every CRSP stock - including the failures - so they carry no survivorship bias. The rule is Moskowitz and Grinblatt's industry momentum: rank the ten industries on their t-12 to t-2 return, go long the best three and short the worst three, equal-weighted, hold one month. That gives 746 months from July 1964.



Takeaway. On survivor-free data momentum is priced: 3.8% a year (t = 2.2), a 4.0% CAPM alpha (t = 2.3) and a 4.9% FF3 alpha (t = 2.8), correlated 0.72 with stock-level momentum. Our decade's t is 0.98, a lack of evidence rather than of premium: a ten-year mean has been positive in 97% of windows since 1974, and at a Sharpe of 0.28 the mean needs 52 years to reach t = 2. Momentum has also been weak since 2003, and costs finish it: 82% monthly turnover, -1.1% a year at 50 basis points, break-even 38 basis points. Priced, but not tradable at this cost.

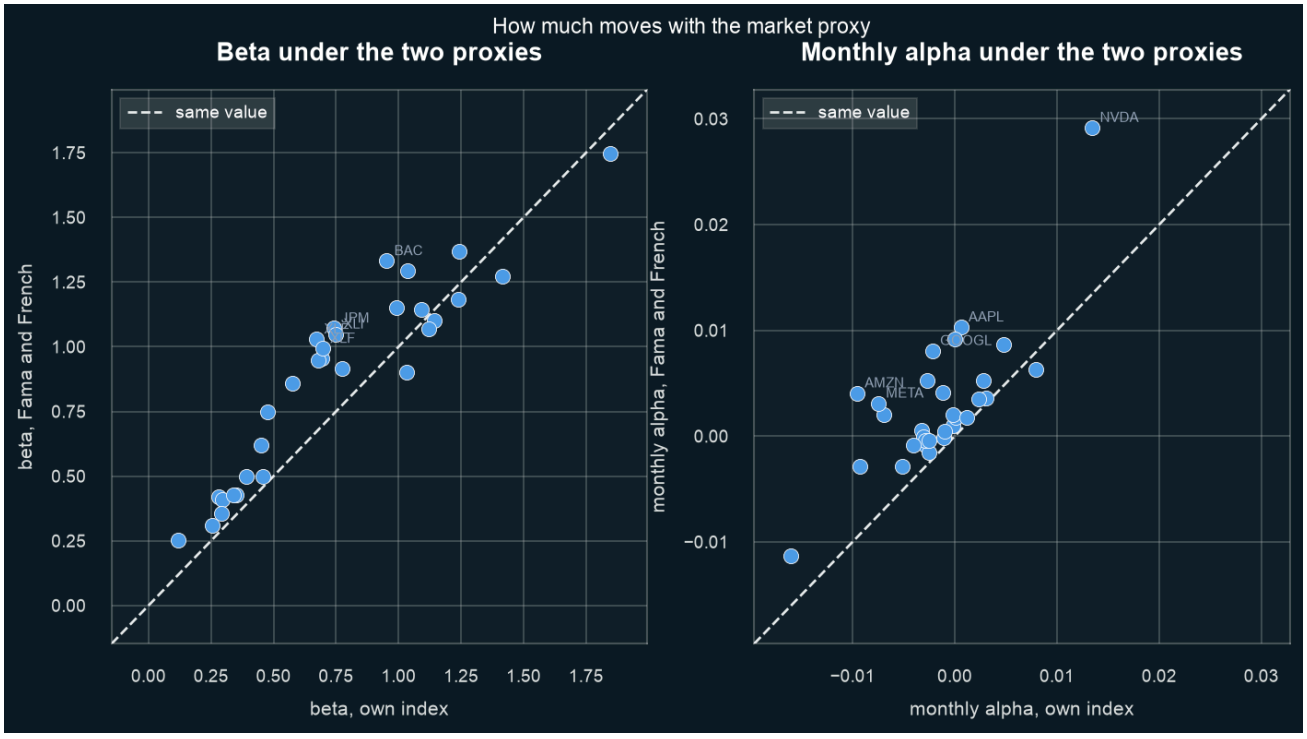
Roll's critique: we test the proxy, not the CAPM

A CAPM test uses a stand-in for the market, and the model requires that stand-in to be mean-variance efficient. Our value-weighted index and the Fama-French market factor are different portfolios of the same market, and swapping them moves

the evidence.

Betas and joint tests under two market proxies. Stage 2 re-estimated with our value-weighted composite and with the Fama and French market factor.

How much moves with the market proxy. Betas (left) and alphas (right) estimated with our value-weighted index against the Fama and French market factor; distance from the dashed line is how much the estimate depends on the proxy.



Takeaway. Switching proxy halves the US GRS p value (0.47 to 0.19), lifts average beta from 0.75 to 0.88 (individual names by up to 0.38), and changes which names look mispriced (DIS and XLY under the index, NVDA under the Fama-French factor). The fitted SML flips from too flat to too steep because the theoretical slope falls from 0.019 to 0.011. The headline statistic is partly a choice the analyst made: Roll's critique in our own results.

The result we trust least

Everything computed on our 30 US names, starting with the comfortable US CAPM pass. Survivorship has removed the failed firms, so the cross-section is cleaner than a live investor's and the self-built factor spreads have no true loser leg.

The result we trust most uses no survivors: the industry momentum estimate on Ken French's industries (3.8% a year, t = 2.2), priced on 62 years of all CRSP stocks and invisible on our 30 names. The universe decides much of the answer, and this universe cannot answer the question.

Appendix

Supplementary exhibits and technical documentation; the main report refers to these by letter.

A. Contribution statement

The signed contribution table appears on the first page. Contributions were roughly equal, and all members reviewed the final report and can defend every number in it.

Signed: Rahul Thaker · Yi-Kai Tsai · Aryan Sethi · Deepank Dewan

B. Use of AI tools

AI models including Claude, ChatGPT and Deepseek were used for assistance with code generation and debugging. Every number, citation and claim in this report was checked by us, and we take responsibility for it.

C. Data sources

- **US prices** - Yahoo Finance adjusted closes (splits and dividends), a total-return proxy, audited against raw closes and dividends (`scripts/audit_yahoo_prices.py`).
- **Singapore prices and corporate actions** - SGX historical prices and corporate-action notices, with Yahoo daily closes and dividends; rebuilt by `goldman.data.cleaning` . Details in `data/cleaning_report_sg.txt` and `data/corporate_actions_sg.csv` .
- **Risk-free rates** - US: the Fama-French `RF` column. Singapore: the month-over-month change in the MAS compounded SORA Index (`data/rf_sg_monthly.csv`).
- **Factors and long-run benchmarks** - Ken French Data Library: FF3 and `RF` , momentum (`Mom`), and the 10 value-weighted US industry portfolios.
- **FX** - Yahoo `SGD=X` month-end USD/SGD, for the USD-quoted Singapore listings.
- **Share counts** - Yahoo Finance and company filings, for point-in-time market caps.

D. Data cleaning and construction

We assemble two ten-year monthly panels from the committed raw files, and the cleaning - not the regressions - is where most of the data work sits. Each universe holds 120 month-end prices, i.e. 119 monthly returns from October 2016 to August 2026: 30 US names (21 mega-caps plus 9 SPDR sector ETFs), and 30 SGX listings of which 9CI is dropped for a 60-month history, leaving 29. Both panels are today's members backfilled, so they carry the survivorship bias that the report treats as its main finding: failed firms are missing, average returns are biased upward, the index is the survivors' own portfolio, and the tests describe a sample selected for having survived.

Point-in-time value weights. The market benchmark is a value-weighted portfolio, so its weights must not see the future. For each month we compute `market cap(t) = split-unadjusted close(t) x shares outstanding(t)` , forward-fill shares from their report dates only, and carry the earliest known count back where a provider has no early history. Yahoo's own `Close` is retroactively divided by every later split, so we first undo each split to recover the true price. The weight panel is long (`asset, date, market_cap, weight`), and every month uses only a share count that was known by then.

US prices. Yahoo adjusted closes (splits and dividends), used as a total-return proxy and audited by rebuilding each series from raw closes plus dividends: the two agree within 0.11% in every month, with no mis-adjusted ex-date.

Risk-free rate. The US rate is the Fama-French `RF` column. For Singapore we build the rate ourselves from MAS's daily "Domestic Interest Rates" export (the raw download is committed at `data/raw/mas/domestic_interest_rates.csv`): after reconstructing dates from the export's split year/month/day columns, the exact one-month risk-free return is the month-over-month change in the compounded SORA Index, `RF_t = SORA_Index(month-end_t) / SORA_Index(month-end_{t-1}) - 1` , which compounds the daily overnight SORA at actual/365 (we checked this against the export's daily quotes to ~1e-11). We deliberately do not average the annualised daily quotes and divide by 12 - that first-order approximation is not exact - so the compounded index is used directly. The resulting monthly series (2005-08 to 2026-09, `data/rf_sg_monthly.csv`) is committed with its provenance in `data/rf_sg_monthly.source.txt` ; if the file is ever missing the loader warns and falls back to a documented constant rather than silently borrowing the US rate.

Singapore - a rebuild, not a download. Yahoo's adjusted close is wrong for several SGX names, so we do not use it. For each ticker we follow Yahoo's unadjusted daily close and dividends next to SGX's own last-traded price, and rebuild the total-return series and caps ourselves:

- *Wrong dividend currency.* ThaiBev (Y92) had every pre-2024 dividend adjusted using the baht amount as if it were SGD, about 26x too large, so its Yahoo series grows 287x over the decade where the true total return is 0.67x. Jardine C&C (C07) adjusts with the US-dollar amount, so each dividend counts at about 78% of its SGD value.
- *Distributions in specie.* Yahoo scales the price history before a spin-off but leaves earlier dividends unscaled, inflating pre-event yields by up to 2x. For the three large spin-offs (U96 Sep-2020, BS6 Apr-2022, BN4 Feb-2023) it also turns the ex-date into a +13% to +22% jump where holders actually gained +3% to +10%.
- *Rebuilt returns.* We take daily `(close + dividend) / previous close` from the split-adjusted close, rescale each pre-event dividend onto the close's basis, and replace each large spin-off's ex-date return with what a holder actually earned: `(SGX ex price + cash dividend + ratio x distributed shares at their SGX price) / SGX cum price` . The smaller distributions (C09, BN4, 9CI) keep Yahoo's price factor, whose error is a few basis points, but their earlier dividends are rescaled.

- *Point-in-time caps.* Caps use SGX's unadjusted month-end price, because Yahoo's scaled close halves Sembcorp's and Yangzijiang's pre-spin caps. Two share-count glitches are overridden - UOB in 2017-03 (the count drops to 0.7% of its neighbours) and Jardine Matheson from 2019-07 to 2022-05 (Yahoo counts the Jardine Strategic cross-holding) - by carrying the last good count across the window.
- *Currency.* DFI Retail, Hongkong Land and Jardine Matheson trade in USD on SGX; their returns and caps are converted at month-end USD/SGD, so every Singapore series is an SGD total return.

Every correction is one row of `data/corporate_actions_sg.csv` with its SGX corporate-action number and source. The rebuild runs offline and repeatably from the committed daily cache, and a test checks that the committed price and weight files reproduce from that cache exactly. `data/cleaning_report_sg.txt` records what each correction changed, ticker by ticker and event by event.

E. Singapore corrections

Growth of one dollar over the sample in Yahoo's adjusted close and in our rebuilt series, local currency, for the five names we corrected.

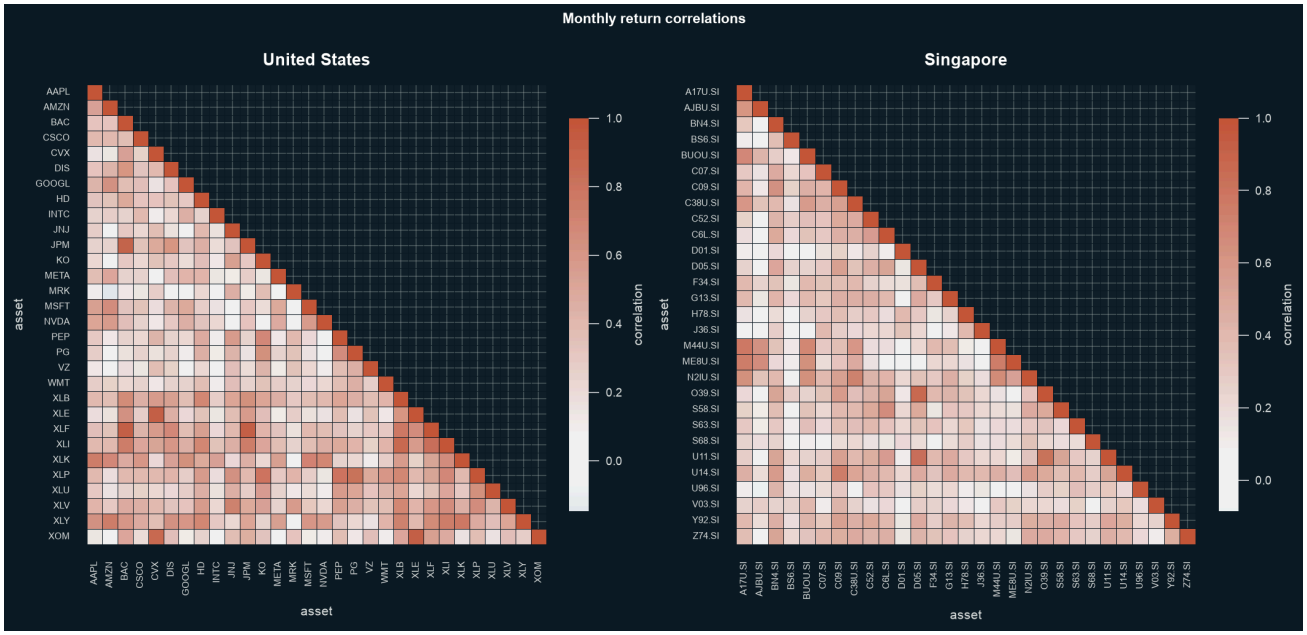
Takeaway. None of the five errors is cosmetic. Yahoo's ThaiBev series compounds 287 times, while its holders actually lost a third of their money (true growth 0.67x). Each of the three large spin-offs leaves one month wrong by 6 to 13 percentage points, which alone moves that stock's average return by 0.6 to 1.4 percentage points a year, and the error sits in the momentum signal for the next eleven months. Every Singapore result below uses the rebuilt series.

	problem	Yahoo growth	rebuilt growth	largest monthly gap (return)
Y92.SI	ThaiBev: baht dividends read as SGD	286.683	0.665	1.591
U96.SI	Sembcorp Ind: Sembcorp Marine demerger, 2020-09	6.338	5.333	0.134
BS6.SI	Yangzijiang: financial arm spin-off, 2022-04	20.830	16.184	0.101
BN4.SI	Keppel: Sembcorp Marine distribution, 2023-02	5.729	4.743	0.060
C07.SI	Jardine C&C: US-dollar dividends read as SGD	0.886	0.937	0.010

F. Correlation structure

How the assets move together, one market per panel.

Takeaway. Each panel is one equity block rather than 30 independent bets: the average pairwise correlation is 0.36 in the US and 0.30 in Singapore, and the tightest pairs are near-substitutes (XLE/XOM at 0.94, BAC/XLF at 0.94, DBS/OCBC at 0.86). A covariance matrix with that much common movement is near-singular in exactly the directions an optimiser wants to trade, which is where Stage 1's extreme weights come from.



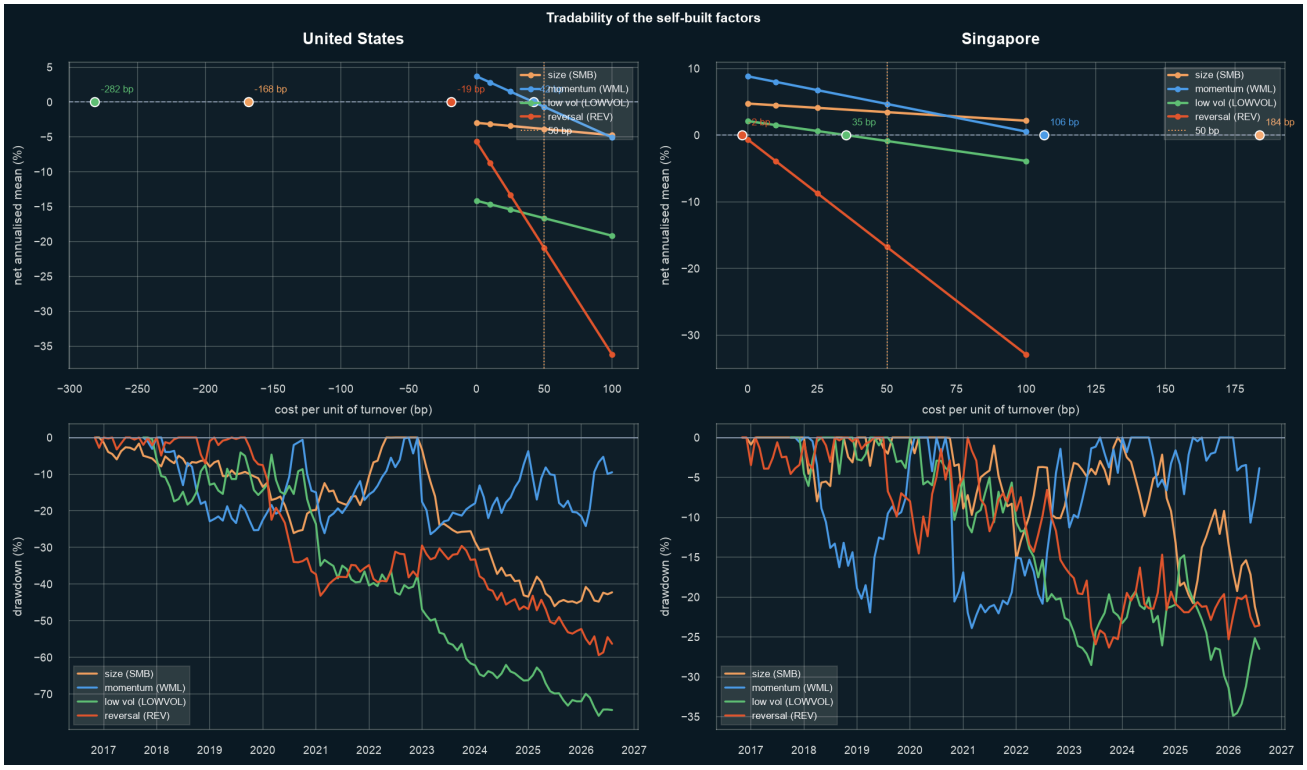
G. Weight stability across markets

	assets	largest absolute tangency weight	half sample weight correlation
us	30.0	2.797	0.354
sg	29.0	0.857	-0.010

Takeaway. Singapore is worse (correlation -0.01), but the more important point is that it fails the same way. Two markets with different currencies, sectors and index construction produce the same collapse, which points at the method rather than at either sample. Regularisation fixes the extremes, but neither panel gives the optimiser the failed firms that would make a short cheap.

H. Factor trading costs

A spread earns marks on paper, not in a portfolio. Each factor rebalances every month and the returns above are gross of the trading it requires. We rebuild each factor's holdings from the same signals, measure turnover as the sum of absolute weight changes (the first month counts the full gross position), and charge a per-unit cost on it. The plot shows the annualised net mean against the cost and marks each factor's break-even cost; the bottom row shows drawdowns. Because the charge is linear in turnover, each line is straight and crosses zero at the break-even cost.



Takeaway. Cost sorts the factors more than the market does. In the US, momentum earns 3.7% a year gross, breaks even at 42 basis points, and keeps -0.7% at 50; size loses 3.0% a year before costs; low volatility loses 14.1%; reversal loses 5.7% and turns over 2.5 times its gross position every month, so it does not break even at any positive cost. In Singapore, size earns 4.7% gross and keeps 3.4% at 50 basis points; momentum earns 8.8% and keeps 4.7%; low volatility earns 2.1% and breaks even at 35 basis points; and reversal loses 0.7% a year gross. The bottom panels are the other half of the story: every factor's peak-to-trough loss is at least 23%, and US low volatility falls 76%.

I. Inference checks for individual alphas

The individual stars above rest on ordinary least squares standard errors, which assume residual variance is constant and residuals are uncorrelated across months. We keep each model's returns, factors and dates fixed and change only the standard errors, using Newey and West (1987), *Econometrica* 55, 703-708. HAC allows changing residual volatility and serial dependence across months. Three monthly lags is our baseline; zero and six are sensitivity checks, not optimised choices, and no lag is selected as the best one.

Individual alphas significant at 5%. OLS uses ordinary standard errors; HAC(0) allows changing volatility only; HAC(3) and HAC(6) also allow serial dependence. Compare across a row: Stage 2 uses the local index; Stage 3 uses the Fama-French market factor for the US and the local index for Singapore.

Stage	Market	Model	Months	OLS	HAC(0)	HAC(3)	HAC(6)
Stage 2	**US**	**CAPM (local index)**	119	2	2	2	2
Stage 3	**US**	**CAPM**	106	1	1	1	3
		FF3	106	1	2	2	4
		FF3 + momentum	106	1	1	2	2
		Self-built SMB + WML	106	1	1	1	2
		Self-built four	106	2	3	5	6
Stage 2	**SG**	**CAPM (local index)**	119	3	4	4	4
Stage 3	**SG**	**CAPM**	107	5	4	3	5
		Self-built SMB + WML	107	5	5	3	6
		Self-built four	107	8	7	8	8

Takeaway. The counts are stable enough to trust the joint tests and fragile enough to distrust single stars. In Singapore's Stage 3 models the significant counts run 5 to 8 under OLS, 3 to 8 under HAC(3) and 5 to 8 under HAC(6); adding the four factors does not consistently reduce them. Stage 2 is more stable (2 of 30 in the US at every lag; 3 to 4 of 29 in Singapore), but HAC turns a non-significant Singapore name, S58, into a significant one at three lags, because classical standard errors understate the precision of its alpha. The sharpest single case is NVDA: it is significant in four of the five US Stage 3 models at three lags and in only two at six. The message for the committee is not to read a count of stars as a list of opportunities; the joint tests and the flat SMLs are the more robust evidence.

Limits. With 106-119 monthly observations, HAC p-values are approximations and are not adjusted for multiple testing. These individual tests neither replace the joint GRS test nor establish trading profits. Alphas and portfolio returns are unchanged; data errors, benchmark timing and survivorship bias remain outside what this check can fix.

J. References

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K. Reproducibility

The notebook is both the report and the entry point. With the committed `data/` files, `uv sync --extra notebook` and then running `notebooks/report.ipynb` top to bottom regenerates every table and figure under `outputs/`; `uv run pytest` checks the pipeline. Package versions are pinned in `uv.lock`, and each inference run records its data hashes and package versions in `outputs/inference_robustness/manifest.json`.